

Your grade: 100%

Your latest: 100% • Your highest: 100% • To pass you need at least 80%. We keep your latest score.

Next item →

1.

Let  $\mathcal{R}$  be a binary relation on a set  $A$ .  $\mathcal{R}$  is said to be a partial order if and only if:

1 / 1 point

☐ it is symmetric and transitive

☒ it is reflexive, anti-symmetric and transitive

☐ it is reflexive and symmetric

☐ it is reflexive, symmetric and transitive



Nice work

Correct. This is the definition of a partial order.
2.

Let  $A = \{a, b\}$  and let  $\mathcal{R}$  be a binary relation on a set  $A$  defined by  $\mathcal{R} = \{(a, a), (a, b), (b, b)\}$ . Which one the following statements is correct?

1 / 1 point

☐  $\mathcal{R}$  is not a partial order as it is not transitive

☒  $\mathcal{R}$  is a partial order

☐  $\mathcal{R}$  is not a partial order as it is not anti-symmetric

☐  $\mathcal{R}$  is not a partial order as it is not reflexive



Nice work

Correct.  $\mathcal{R}$  is reflexive, anti-symmetric and transitive.
3.

Let  $A = \{a, b, c\}$  and let  $\mathcal{R}$  be a binary relation on a set  $A$  defined by  $\mathcal{R} = \{(a, a), (b, b), (a, b), (b, a), (c, c)\}$ . Which one the following statements is correct?

1 / 1 point

☐  $\mathcal{R}$  is a partial order

☐  $\mathcal{R}$  is not a partial order as it is not transitive

☐  $\mathcal{R}$  is not a partial order as it is not reflexive

☒  $\mathcal{R}$  is not a partial order as it is not anti-symmetric



Nice work

Incorrect.  $\mathcal{R}$  is n not anti-symmetric as  $a\mathcal{R}b$  and  $b\mathcal{R}a$  but  $a \neq b$ .
4.

Let  $\mathcal{R}$ , a binary relation on  $\mathbb{Z}$  (a set of all integers), be defined as follows  $a\mathcal{R}b$  if and only if  $a < b, \forall a, b \in \mathbb{Z}$ . Which one of the following statements is true about the relation  $\mathcal{R}$ ?

1 / 1 point

☐  $\mathcal{R}$  is not a partial order as it is not anti-symmetric

☐  $\mathcal{R}$  is a partial order

☒  $\mathcal{R}$  is not a partial order as it is not reflexive

☐  $\mathcal{R}$  is not a partial order as it is not transitive



Nice work

Correct.  $\mathcal{R}$  is not reflexive because no element  $a \in \mathbb{Z}$  is related to itself as  $a \not\prec a$ .
5.

Let  $\mathcal{R}$ , a binary relation on  $\mathbb{Z}$  (a set of all integers), be defined as follows  $a\mathcal{R}b$  if and only if  $a \bmod 2 = b \bmod 2, \forall a, b \in \mathbb{Z}$ . Which one of the following statements is true about the relation  $\mathcal{R}$ ?

1 / 1 point

☒  $\mathcal{R}$  is not a partial order as it is not anti-symmetric

☐  $\mathcal{R}$  is not a partial order as it is not transitive

☐  $\mathcal{R}$  is a partial order

☐  $\mathcal{R}$  is not a partial order as it is not reflexive



Nice work

Correct. It is not anti-symmetric as it doesn't satisfies the anti-symmetry property. For example:  $\mathcal{R}$   $2\mathcal{R}4$  and  $4\mathcal{R}2$  but  $2 \neq 4$ .
6.

Let  $A = \{a, b, c\}$  and let  $\mathcal{R}$  be a binary relation on a set  $A$  defined by  $\mathcal{R} = \{(a, a), (b, b), (c, c)\}$ . Which one the following statements is correct?

1 / 1 point

☒  $\mathcal{R}$  is a partial order but is not a total order

☐  $\mathcal{R}$  is a total order relation

☐  $\mathcal{R}$  is not a total order as it is not transitive

☐  $\mathcal{R}$  is not a total order as it is not a partial order



Nice work

Correct.  $\mathcal{R}$  is a partial order as it is reflexive, anti-symmetric and transitive. However,  $\mathcal{R}$  is not a total order as  $a\not\mathcal{R}b$  and  $b\not\mathcal{R}a$ .